

ITAI 1371 – Intro To Machine Learning

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Lab 05 Reflective Journal – Data Preparation

This lab helped us understand why data preparation is such an important part of the machine learning workflow. In the previous lab, we used Exploratory Data Analysis to identify patterns and problems in the Titanic dataset. In this lab, we moved one step further and started making decisions about how to prepare that data so that it can eventually be used by a machine learning model.

One of the first concepts that became clearer to us was missing value imputation. The Age column originally contained 177 missing values. Instead of deleting all of those passengers, we used the median age to replace the missing information. After the imputation, the Age column had zero missing values. This helped us understand that cleaning data does not always mean removing information. Sometimes we can preserve useful records by replacing missing values with a reasonable estimate.

We also understood better why the median can sometimes be more appropriate than the mean. If a numerical feature contains extreme values or has a skewed distribution, those values can pull the mean higher or lower. The median represents the middle of the distribution and is less affected by outliers. This made us realize that choosing how to fill missing information should not be automatic. The strategy should depend on the characteristics of the data.

Another important concept in this lab was One-Hot Encoding. The Titanic dataset contains categorical information such as Sex and Embarked. Machine learning models generally need numerical information, so those text categories cannot simply be passed to many algorithms in their original form. Using `pd.get_dummies()` transformed the categories into new binary features such as `Sex_male``, `Embarked_Q``, and `Embarked_S``.

Before this lab, converting categories into numbers sounded simple, but we now understand that the method matters. If we assigned arbitrary numbers such as 1, 2, and 3 to categories that do not have a natural order, a model could incorrectly interpret those numbers as a ranking. One-Hot Encoding avoids creating that artificial relationship. We also used `drop_first=True``, which showed us that we can avoid unnecessary redundant columns while still representing the categorical information.

Feature scaling was another concept that became more understandable through this exercise. Age and Fare originally had very different ranges. Fare could contain values much larger than Age, even though a larger numeric range does not necessarily mean that a feature should be more important. Using `StandardScaler` transformed both features to a common scale. Seeing the original values change into positive and negative standardized values helped us understand that scaling changes the representation of a feature, but it does not change what that feature means.

We also learned that feature scaling is not required for every machine learning algorithm. For example, Decision Trees compare individual features using threshold-based splits, so they are not affected by whether one feature ranges from 0 to 80 and another ranges from 0 to 500. This was important because it showed us that data preparation is not a fixed checklist where every transformation must always be applied. The preprocessing decisions depend on the type of data and the model we plan to use.

Something else we noticed is that this lab focused on preparing specific parts of the Titanic dataset rather than completely cleaning every column. For example, the original dataset also contains many missing values in Cabin and a small number in Embarked. This made us think more carefully about the idea that different variables may require different strategies. A column with only a few missing values might be handled differently from a column where most of the information is missing.

The biggest lesson from this lab is that data preparation creates the connection between understanding the data and actually building a machine learning model. EDA helps us discover the problems, while data preparation requires us to decide what to do about them. Missing values, categorical features, and numerical scales can all influence how successfully a model can use the information.

This lab also helped us understand the requirements for our Midterm project more clearly. We now see why choosing a dataset that already contains only clean numerical values would remove an important part of the machine learning process. A dataset with missing information, categorical features, different numerical scales, and other imperfections gives us the opportunity to demonstrate EDA, cleaning, encoding, scaling, and preparation before model training.

Overall, this lab made the preprocessing stage feel much more logical. Instead of seeing cleaning, encoding, and scaling as separate Python commands, we now understand the reason behind each transformation and how those decisions prepare real-world data for machine learning.